STM32F3 Technical Training

For reference only

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STM32F37x Specific Features/ peripherals

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Sigma delta analog to digital converter (SDADC)

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SDADC introduction (1/2)

- Sigma delta principle inside STM32:
  - High precision (new applications: medical, metering, gaming)
  - Excellent linearity (simplifies calibration)
  - No sample & hold

- Main properties:
  - 3 Σ-Δ ADCs in all packages (19 single ended and 10 differential inputs max.)
  - 16-bit resolution, ENOB = 14 bits (SNR = 89dB)
  - Low power modes:
    - Slow (speed reduced 4x): up to 600uA (instead of 1200uA in run mode)
    - Standby: up to 200uA, wakeup time 50us
    - Power down: up to 10uA, wake up time 100us
  - Internal or external reference voltage usage
  - Independent power supply pins: SDADCx_VDD
  - Conversion rates:
    - Up to 50ksps in fast mode (single channel)
    - Up to 16.6ksps in normal mode (multiple channels)
  - 7 programmable gains: ½, 1, 2, 4, 8, 16^*, 32^* (* = digital gains)

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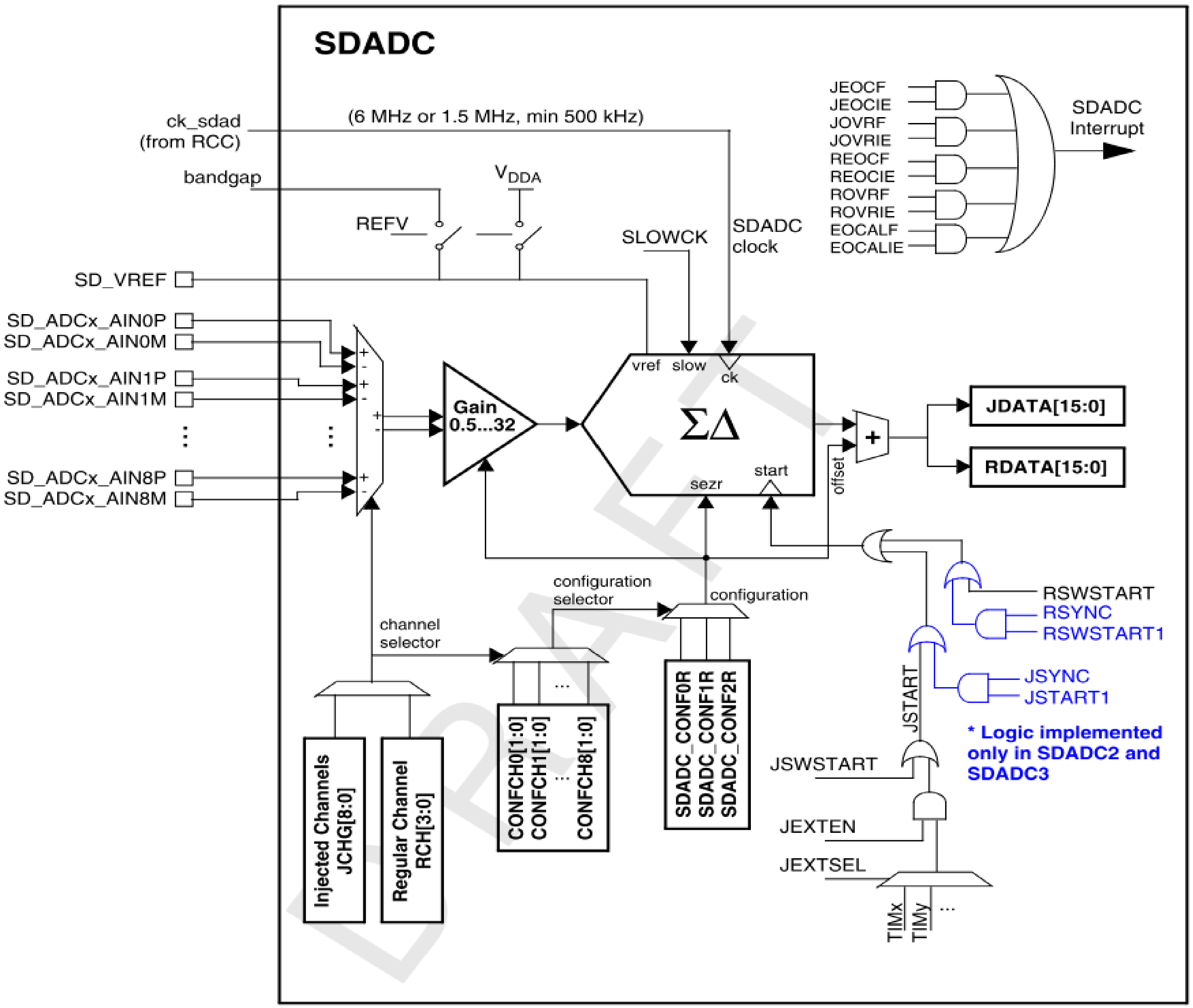
SDADC introduction (2/2)

- Next features:
  - 9 single ended inputs or 5 differential inputs per one SDADC (or combination)
  - DMA capability to transfer data to RAM (conversion when CPU in sleep mode)
  - Triggers:
    - Software
    - Timer
    - External pin
    - Synchronization to first SDADC (SDADC1)
  - Signed output data format (16-bit signed number)
  - Zero offset calibration
  - 3 measuring modes – per analog channel selection:
    - Single ended referenced to zero
    - Single ended offset mode
    - Differential mode
  - Interrupts and flags:
    - Interrupts: EOCAL, REOC, JEOC, ROVR, JOVR
    - Flags: STABIP, CALIBIP, RCIP, JCIP

SDADC block diagram

- One SDADC configuration:



SDADC pins

Name	Signal type	Remarks
SDADCx_VDD	Input, analog Supply	Analog power supply. Must be greater than 2.4 V (or 2.2 V in Slow mode) and less than 3.6 V.
SDADCx_VSS	Input, analog supply ground	Analog ground power supply.
SDADCx_AIN[8:0]P	Analog input	Positive differential analog inputs for the 9 channels
SDADCx_AIN[8:0]M	Analog input	Negative differential analog inputs for the 9 channels.
SD_VREF+	Input or In/Out, positive analog Reference	When the external reference is selected (REFV=00), this pin must be driven externally to a voltage between 1.1 V and SDADCxVDD (minimum for x=1..3).When an internal reference is selected (REFV is 01, 10, or 11), this pin must have an external capacitance connected to SD_VREF-
SD_VREF-	Input, negative analog reference	This pin, when present, must be driven to the same voltage level as SDADCxVSS.

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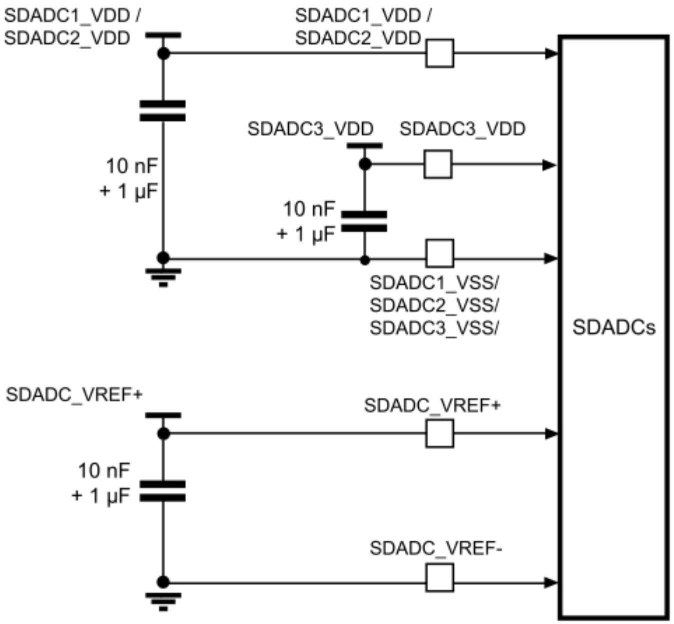
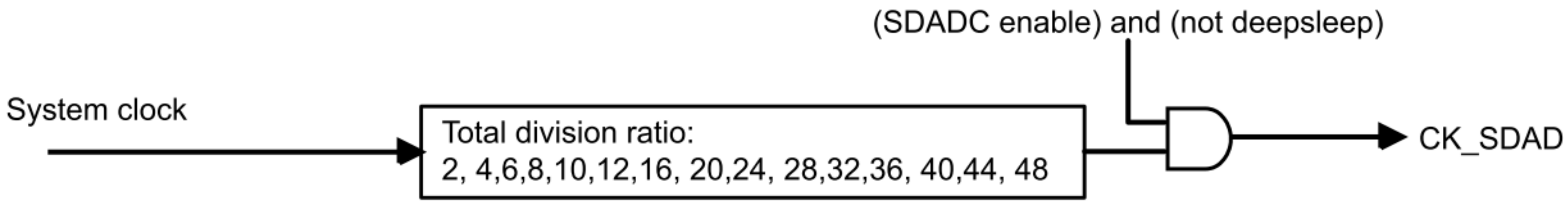
SDADC power supply and reference voltages

- Power supply:
  - Independent power supplies:
    - SDADC1/2 _VDD– for SDADC1 and SDADC2
    - SDADC3_VDD – for SDADC3
    - SDADCx_VSS – common for all SDADCs
  - Voltage range:
    - Full speed mode operation: 2.4V – 3.6V
    - Slow mode operation: 2.2V – 3.6V



- Reference voltage selection:
  - Internal:
    - Internal bandgap voltage: 1.2V
    - Internal bandgap voltage amplified by 1.5x : 1.8V
    - VDDA power supply
  - External
    - Dedicated SDADC_VREF+ , SDADC_VREF- pins
    - Voltage range 1.1V – SDADCx_VDD

SDADC clock



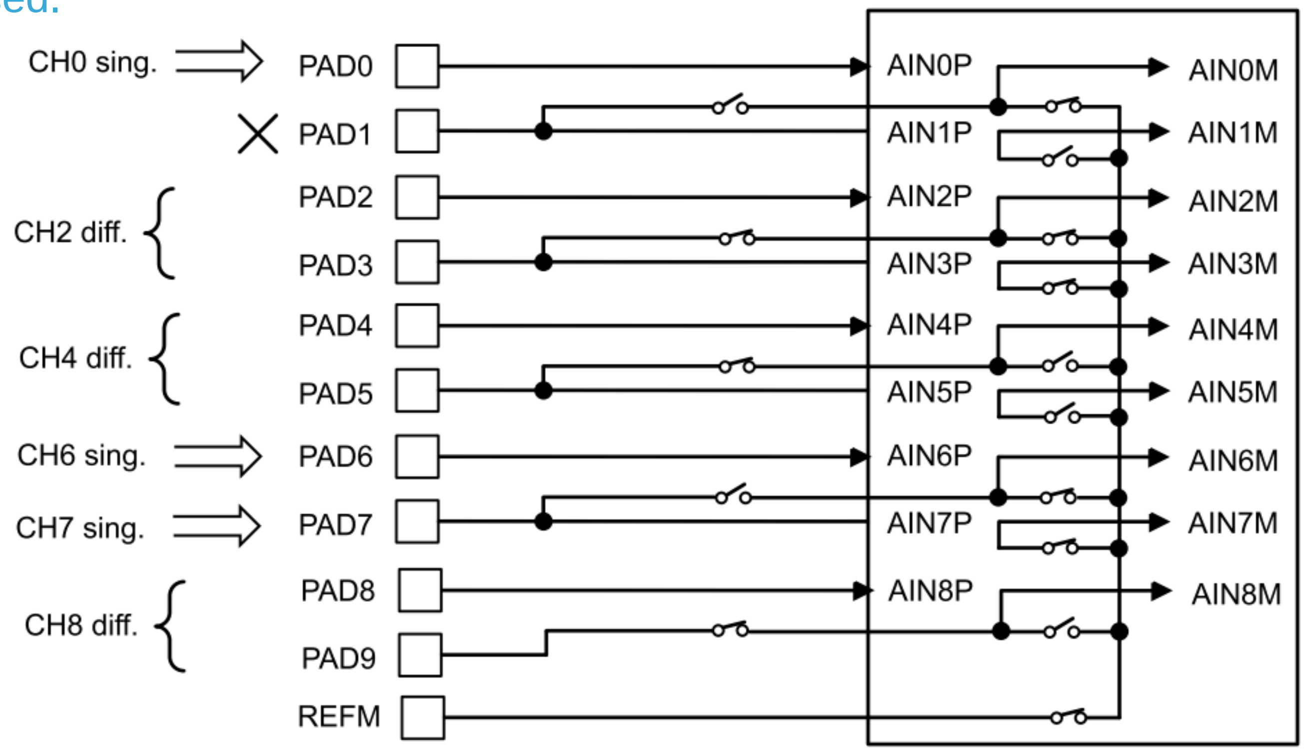
- Clock management:
  - System clock divided by divider (from 2 to 48, 50% duty cycle)
  - Clock range:
    - max. 6MHz – standard conversion clock
    - max. slow mode clock 1.5MHz – reduced speed, reduced power, lower voltage operation
    - min. clock speed = 500kHz

Input channel configurations

- Measurement modes:
  - Differential mode:
    - Used both SDADC analog channel inputs: SDADCx_AINxP and SDADCx_AINxM
    - Signed result: 0x8000 – 0x7FFF (-32768 – 32767)
  - Single ended modes:
    - Offset mode: as differential mode with minus input internally grounded (reduced dynamic range of SDADC – only positive range: 0x0000 – 0x7FFF)
    - Referenced to zero: minus input internally grounded but offset injected to have full dynamic range (zero voltage corresponds to code -32768)
- Three SDADC configuration registers (SDADC_CONFxR, x = 0..2) => 3 possible configurations:
    - In each register is channel configuration:
      - Measurement mode (differential or single ended)
      - Gain (½ , 1, 2, 4, 8, 16, 32)
      - Offset calibration value (stored here after offset calibration)
      - Common voltage used during offset calibration (VSSA, VDDA, VDDA/2)
  - Each SDADC analog channel is assigned to one configuration register
    - Example: 3 analog channels in application
      - Channel 0 uses SDADC_CONF0R
      - Channel 1 and channel 2 use SDADC_CONF1R (same gain and measuring mode)

Channels configuration example

- Mixed configurations – example of input pins connection:
  - CH2, CH4 and CH8 are used as differential.
  - CH0, CH6 and CH7 are used in single-ended mode.
  - REFM is used – VSSA.
  - PAD 1 is not used.



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Regular and injected conversions

- Injected conversions
  - Injected group is defined as bitfield in register – each one bit corresponds to one channel
  - Selected channels in the injected group are always converted sequentially (from lowest selected channel) – scan mode
  - Triggers:
    - Software (writing ‘1’ to the JSWSTART bit)
    - External pin
    - Timers
    - Synchronous with SDADC1
- Regular conversions
  - Channel selection is defined as channel number in register
  - Cannot run in scan mode
  - Triggers:
    - Software (writing ‘1’ to the RSWSTART bit)
    - Synchronous with SDADC1

Standard, slow, low power conversion modes

- Standard mode:
  - Normal:
    - Multiplexing more channels
    - One conversion takes 360 cycles (16.6ksps @ 6MHz)
  - Fast continuous (FAST = 1):
    - On one channel only in continuous mode – regular channel or one injected channel selected
    - One conversion takes 120 cycles (50ksps @ 6MHz)
- Slow mode (SLOWCK = 1):
  - Reduced power consumption (~600uA consumption), operation from 2.2V
  - Limited clock speed – up to 1.5MHz (so 4x reduced also conversion rate)
- Standby when idle (SBI = 1):
  - SDADC goes to standby when no conversion (~200uA consumption)
  - Needed time for wakeup from power down – 50us
- Power down when idle (PDI = 1):
  - SDADC goes to power down when no conversion (~10uA consumption)
  - Needed time for wakeup from power down – 100us

Request precedence

- Priority order of SDADC operations:
    1. Calibration sequence
    2. Injected conversions
    3. Regular conversions

- But:
    - Conversion which is already in progress is never interrupted by the request for another action (current conversion is finished first)
    - Request is ignored if a like action is already pending or in progress
    - No action can start before stabilization has finished (wakeup from power down or standby mode)

SDADC calibration

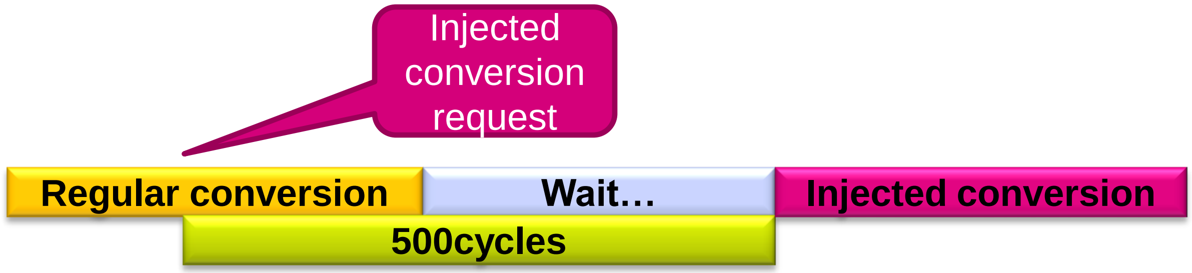
- General properties for sigma delta converters:
  - Perfect linearity (due to 1-bit converter and oversampling)
  - Resolution increases with decreasing data rate
  - But large offset and gain error (need calibration)

- Offset calibration:
  - Principle:
    - Short internally both channel inputs (positive and negative)
    - Perform conversion and store result to configuration register(s)
    - During standard conversion subtract from result the calibrated value
  - Implementation in STM32F37x:
    - Set in configuration registers:
      - required gain (1/2 .. 32)
      - common mode for calibration (VSSA, VDDA, VDDA/2)
    - Set how many configurations to calibrate (CALIBCNT[1:0] bits)
    - Start calibration by setting bit STARTCALIB
    - Calibration sequence then executes on given gain(s) :
      - Calibration values are stored into configuration registers (OFFSETx[11:0] bits)
      - 30720 cycles (5.12 ms at 6 MHz) for one configuration register
    - Calibration data are automatically subtracted from each conversion data

Deterministic timing

- Application requirements:
  - Launching conversion in precise intervals (e.g. FFT sampling by timer trigger)
  - Problem: waiting for some ongoing (regular) conversion

- Solution in SDADC:
  - Start of each injected conversion with delay during which cannot be started regular conversion
  - When bit JDS = 1 (Injected Delay Start) the start of each injected conversion is delayed:
    - by 500 cycles if PDI = 0 (power down when idle)
    - by 600 cycles if PDI = 1, SLOWCK = 0 (because wakeup from power down takes 600 cycles)



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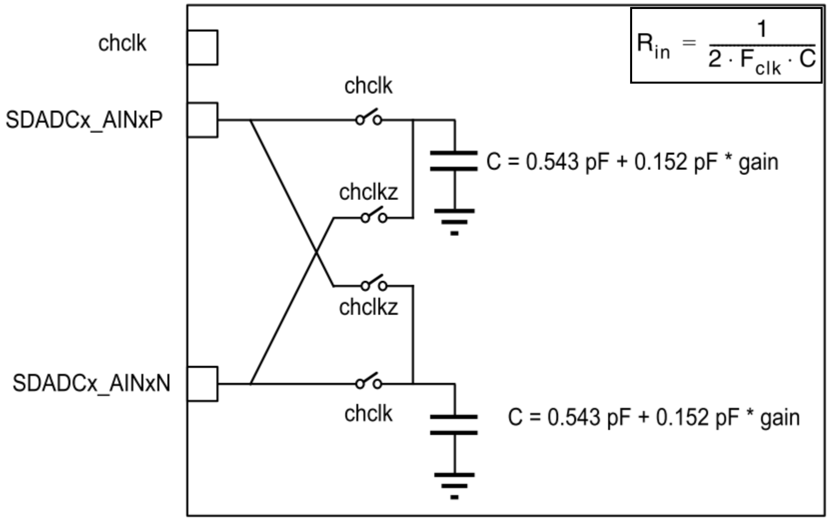
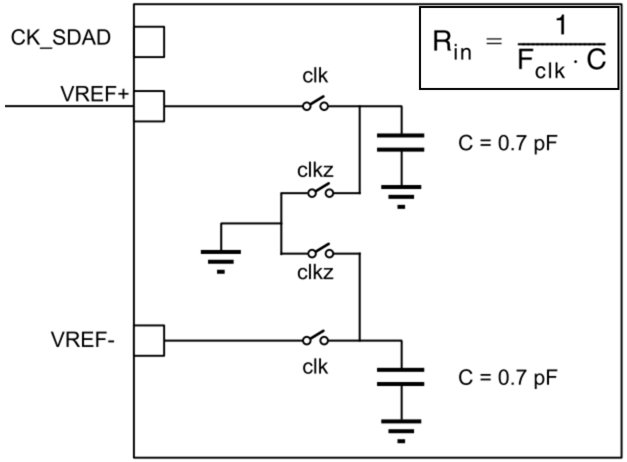
Inputs impedances

- Analog inputs impedance:
  - Depends from:
    - selected SDADC clock
    - analog gain (0.5 – 8)
    - conversion is in progress
  - Switching capacitance character
  - Range (examples):
    - 540kΩ @ 1.5MHz, gain = 0.5
    - 135kΩ @ 6MHz, gain = 1
    - 47kΩ @ 6MHz, gain = 8



- Reference voltage input impedance:
  - Depends only from selected SDADC clock
  - Switching capacitance character
  - Range (6MHz – 1.5MHz):
    - ~ 230kΩ – 1000kΩ



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Quiz

- How many analog input channels are in one SDADC in the STM32F37x ?
- What is the calibration result ?
- What is the conversion modes regarding low power conversions ?
- Which voltages can be used as reference voltage for SDADC ?
- What is the priority regarding conversions order ?

ADC 1 MSPS

ADC Features

- Same like in STM32F1 family:
  - 12-bit, 1Msps
  - Triggers, self-calibration
  - Up to 18 input analog channels
  - Analog watchdog, interrupts, DMA
  - Programmable sampling time, Vref+ input range
  - Injected, regular channels, alignment
  - Continuous, single, scan conversion modes
  - Temperature sensor, Vrefint measuring
- Added feature:
  - VBAT measuring

Thank you !

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